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INTERFACE CONTROL DOCUMENT – CARRIER TO MODULES

Document: ICD-EEG-006 **Revision:** C **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0
Governing documents: DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Revision note. Rev B carries the two geometry findings of the layout work – the carrier is 150.0 x 130.0 mm and it is a four-layer board – regenerates the connector register from tools/design.py at the new coordinates, resizes MP-01 and POD-P1 to match, and closes the Rev A open points on keying, I2C pull-ups, VBUS_DET, ENV_CMP, E-04, E-11 and E-28 against the package v2 rulings.

The findings of the second cross-document audit are closed in this issue, within the same revision letter and without a further letter change: the module count is stated unconditionally, the MP-01 geometry is restated from tools/mech_gen.py, RFQ E-23's 45 C charge inhibit is stated as not met, and the rail budgets of sections 2.7 and 5.3 are reconciled with each other and read against TST-EEG-004 T3. The rulings those corrections apply are RUL-EEG-021 Rev B, which is now a controlled document in docs/.

Corrections within Rev B, after the design and firmware changes of 2026-09-02, made in place with their date and without a further revision letter: **ECO-EEG-024 is applied**, so the series protection resistance a module must tolerate is **68 kOhm**, not 47 kOhm, and RFQ S-02's single-fault DC limit is **met at 36.8 uA** – section 2.1's acceptance row, section 2.1's protection-network paragraph, the section 8 incoming-module step 8 and open item 6 are restated for it, and E-10 now sits on the +/-1.0 dB branch its own wording provides. **The contact-light bicolour phase driver is written**, so section 2.11 and open item 11 no longer say the interface has nothing driving it. The interface itself did not change in either case, and neither change has been measured on hardware: no unit exists.

Why this document exists

EEG-CAR-01 Rev B is a carrier, not an instrument. Twelve purchased module types do the work, and no public standard fixes where any of them puts its header. Rev A.2 drew sockets at guessed coordinates and would have discovered the guess when the boards arrived. Package v2 removes the dependency: every module except the ESP32-S3-DevKitC-1 sits on the printed module plate MP-01 above the carrier and is joined to it by a made-up 2.54 mm ribbon jumper, so the carrier sockets are placed for routing and the jumper absorbs whatever geometry the module turns out to have. That only works if somebody writes down, way by way, what each socket carries and what each module must present. This document is that. It is the reference against which TST-EEG-004 step T2 checks continuity, the source from which the module symbols of the schematic set SCH-EEG-005 are generated, and the form a builder fills in before accepting a substitute part.

There is a KiCad symbol library for this board from Rev C, and it is generated. kicad/EEG-CAR-01.kicad_sym is written by tools/emit_kicad_sch.py from tools/schlib.py, which builds every connector symbol from the conn() calls in

tools/design.py that section 1.1 below is transcribed from – so a connector symbol whose way count or pin names disagree with this document is not possible. Package v2 Rev B said no such file existed and that Rev A was wrong to claim one; **that was true of Rev B and is superseded by ECO-EEG-033**, which added the native schematic kicad/EEG-CAR-01_RevC*.kicad_sch for the same reason. It is not edited by hand. The eight-sheet PDF set SCH-EEG-005 continues to be drawn separately by tools/schematic.py for Rev B and is a picture, not a netlist; the Rev C schematic is the netlist-bearing one.

Nothing in this package has been built or measured, and no safety engineer has reviewed this design. Every derived figure is labelled *calculated* and the first article is where the calculations stop being calculations.

1. The carrier connector register

The carrier is **150.0 x 130.0 mm** and has **four layers**: L1 signal, L2 reference plane, L3 reference plane, L4 signal. Package v1 asserted that two layers would be enough for this board. Doing the layout showed that they are not: on two layers the bottom side has to be both the reference plane and the second routing surface, and it cannot be both. Four layers give two whole routing surfaces and a continuous reference under every analogue trace, which is what the layout rules of **DSN-EEG-002 section 13** – the rules that are requirements and not preferences – ask for, and a swiss-cheesed two-layer pour cannot deliver it. The zoning, the star point and the isolation keep-out are **DSN-EEG-003 section 3.3**. The reference planes are AGND_REF left of $x = 62$ mm and DGND right of it, on **both** inner layers, tied together by stitching vias, so every analogue trace on L1 or L4 has continuous reference copper under it and both routing surfaces stay whole. Vias are **through vias only**, 0.60 mm pad on a 0.30 mm finished hole, no blind, buried, back-drilled, filled or plugged. The stack is mask / 35 μ m L1 / prepreg 0.200 / 17 μ m L2 / core 1.065 / 17 μ m L3 / prepreg 0.200 / 35 μ m L4 / mask = **1.60 mm $\pm$ 10 %**. The full board specification is DSN-EEG-003 section 3.2 and is not restated here.

The routed result against this connector set. kicad/EEG-CAR-01_RevB_DRC_report.txt is the authority for it and every figure below is read from that report. EEG-CAR-01 Rev B is routed on four layers with **3 745 track segments and 552 through vias**. **All 145 nets are fully connected** – none unclosed, none without copper – and each reference plane is one continuous island per net on both inner layers. Every geometric rule passes: the smallest measured clearance is 0.260 mm on F.Cu, 0.275 mm on B.Cu and 0.285 mm on the planes against a 0.20 mm rule; the narrowest conductor is 0.200 mm; the smallest plated hole is 0.300 mm; copper stands 2.00 mm clear of every non-plated hole; no digital net enters the analogue zone; there are no duplicate segments and no duplicate via positions; and there is exactly one AGND_REF-to-DGND bridge and one HARN_SHIELD-to-DGND bridge. The report records **no violation of any kind** – its own line is “VIOLATIONS: 0 – none. The board passes every rule listed above” – so **the isolation strip of DSN-EEG-003 section 3.3 is free of copper on all four layers**, the two inner planes included. That is written here because the report says it, not because it was intended.

The board closes, and it closes at minimum geometry. The same report lists **169 connections the router had to relax**: 36 took a conductor narrower than the 0.25 mm preferred width, and 133 kept full width and took a reduced gap instead. All 169 are at or above the 0.20 mm minimum conductor and the 0.20 mm minimum gap, so all 169 pass;

a board that closes at the minimum is still not the same board as one that closes with margin everywhere, and the relaxed connections are listed pad by pad so a reviewer can see where.

Rev B is WITHDRAWN FROM FABRICATION and Rev C is unrouted. The release gate of ECO-EEG-016 section 3 – zero DRC violations, every net one connected copper island, both inner planes continuous under the analogue zone – was met on all three counts, and it was not enough. An external layout engineer read the Rev B board between 3 and 5 September 2026, declined the review and advised a redesign; his seven findings are ECO-EEG-016 section 2, ECO-EEG-030, and six of them are rules the programme’s DRC was never given. No board is ordered from Rev B and its data stays in the tree as history. **Rev C is the same circuit at corrected footprints and part numbers, with no routing of its own:** placement and routing are bought, and Rev C is released to the layout contractor under RFQ-EEG-002A as re-scoped by ECO-EEG-030, under the rules of LAY-EEG-034. **Nothing in this package is released for fabrication.** Every connector position, pin assignment and net name in this document is unchanged by that: Rev C moves footprints and part numbers only.

Modules to connectors. This table is the one home of the module-to-connector mapping; DESIGN_FACTS, RFQ-EEG-001 and DSN-EEG-003 cite it rather than repeating it.

Block	Module type	Carrier connectors
Analogue front end x2	ADS1299 eight-channel breakout (PIEEG-8 class)	#1: J1, J2, J23. #2: J3, J4, J29. Daisy stub J5
Controller	ESP32-S3-DevKitC-1-N16R8	J6, J7 (direct insertion, 22.86 mm apart)
Audio codec	ES8388-class codec with headphone amplifier	J8, J9
USB isolation	ADuM4160 module, host receptacle on the module	J10
Secure element	ATECC608B breakout	J11
Charger	bq24074-class charger with power path	J12 (with the gauge), J13 cell, J24 charge input
Fuel gauge	MAX17048-class gauge	J12 (with the charger)
Rail	TPS63020-class buck-boost	J25
Storage	microSD breakout, one-bit SDMMC	J20
Boom microphone preamplifier	fixed-gain electret preamplifier, see 2.9	J21, with the boom pigtail at J18
Room microphone	MEMS or electret module with hardware mute	J28
Contact lights	74HC595 shift-register module	J19, with the light harness at J30

Twelve module types and thirteen module assemblies per unit, unconditionally.

The thirteenth assembly is the second ADS1299 breakout: the type is specified once and fitted twice. The charger and the gauge are two of the twelve types and count as two assemblies whichever way they are supplied – the baseline combined charger-plus-gauge board of section 2.6 carries both of them on one J12 jumper, and two separate breakouts carry one each on the Y jumper of section 3.3 – so neither arrangement changes either count. Twelve of the thirteen assemblies mount on MP-01; the thirteenth is the ESP32-S3-DevKitC-1, which is inserted directly into J6 and J7. This is the count of RUL-EEG-021 section B and it is the count used in RFQ-EEG-001, DSN-EEG-003, TST-EEG-004, ASM-EEG-007, AVL-EEG-017, QP-EEG-010, PKG-EEG-015, SVC-EEG-013 and REG-EEG-012.

Coordinates below are the design-source convention: origin at the top-left board corner, X right, Y down, from `tools/design.py`. Pin 1 of every socket strip is the square pad and is marked on the top silkscreen. Way counts are the carrier socket way counts. The **Shroud** column says whether the socket carries a WH-KEY-01 printed keying shroud (section 6.1).

Ref	Footprint	Ways	Position (mm)	Zone	Class	Shroud	What connects
J1	PinSocket_1x12_P2.54mm_Vertical	12	66.0, 6.0	D	module	yes	ADS1299 module #1, digital
J2	PinSocket_1x10_P2.54mm_Vertical	10	41.0, 5.0	A	module	yes	ADS1299 module #1, analogue signals
J23	PinSocket_1x06_P2.54mm_Vertical	6	47.0, 5.0	A	module	yes	ADS1299 module #1, analogue rails
J3	PinSocket_1x12_P2.54mm_Vertical	12	66.0, 42.0	D	module	yes	ADS1299 module #2, digital
J4	PinSocket_1x10_P2.54mm_Vertical	10	41.0, 50.0	A	module	yes	ADS1299 module #2, analogue signals
J29	PinSocket_1x06_P2.54mm_Vertical	6	47.0, 24.0	A	module	yes	ADS1299 module #2, analogue rails
J5	PinSocket_1x04_P2.54mm_Vertical	4	72.0, 6.0	D	module	yes	ADS1299 module #1 DAISY_IN / CLKOUT stub
J6	PinSocket_1x22_P2.54mm_Vertical	22	82.0, 8.0	D	direct	no	ESP32-S3-DevKitC-1 row A
J7	PinSocket_1x22_P2.54mm_Vertical	22	104.86, 8.0	D	direct	no	ESP32-S3-DevKitC-1 row B
J8	PinSocket_1x14_P2.54mm_Vertical	14	90.0, 72.0	D	module	yes	audio codec module
J9	PinSocket_1x04_P2.54mm_Vertical	4	114.0, 62.0	D	module	yes	codec microphone feeds
J10	PinSocket_1x04_P2.54mm_Vertical	4	136.0, 6.0	D	module	yes	USB isolator, device side

Ref	Footprint	Ways	Position (mm)	Zone	Class	Shroud	What connects
J11	PinSocket_1x04_P2.54mm_Vertical	4	136.0, 20.0	D	module	yes	secure element breakout
J12	PinSocket_1x08_P2.54mm_Vertical	8	136.0, 34.0	D	module	yes	charger and fuel gauge
J13	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	2	136.0, 60.0	D	cell	no, JST keyed	protected 18650 carrier
J14	PinSocket_1x12_P2.54mm_Vertical	12	5.0, 12.0	A	harness	yes, WH-EEG-008	electrode cable, 12-way screened
J15	DIN42802_1p5mm_Socket	1	8.0, 76.0	A	panel	no, touch-proof	EMG1 cheek
J16	DIN42802_1p5mm_Socket	1	8.0, 88.0	A	panel	no, touch-proof	EMG2 submental
J17	DIN42802_1p5mm_Socket	1	8.0, 100.0	A	panel	no, touch-proof	EMG3 laryngeal
J18	PinSocket_1x04_P2.54mm_Vertical	4	122.0, 90.0	D	panel	no, marked only	boom microphone TRRS pigtail
J19	PinSocket_1x16_P2.54mm_Vertical	16	78.0, 72.0	D	module	yes	74HC595 contact-light driver
J20	PinSocket_1x08_P2.54mm_Vertical	8	136.0, 72.0	D	module	yes	microSD breakout, one-bit SDMMC
J21	PinSocket_1x06_P2.54mm_Vertical	6	122.0, 72.0	D	module	yes	boom microphone preamplifier
J22	PinSocket_1x03_P2.54mm_Vertical	3	15.7, 73.5 (270 deg)	A	panel	yes, WH-EEG-008	EOG / spare electrode header, Phase 2 option
J24	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	2	143.0, 80.0	D	panel	no, JST keyed	charge-only USB-C pigtail
J25	PinSocket_1x06_P2.54mm_Vertical	6	128.0, 86.0	D	module	yes	buck-boost module
J26	PinSocket_1x06_P2.54mm_Vertical	6	117.0, 8.0	D	debug	no, marked only	console and recovery header
J27	PinSocket_1x04_P2.54mm_Vertical	4	128.0, 72.0	D	panel	no, marked only	3.5 mm headphone jack pigtail
J28	PinSocket_1x04_P2.54mm_Vertical	4	122.0, 102.0	D	module	yes	room microphone module
J30	PinSocket_1x10_P2.54mm_Vertical	10	66.0, 90.0	D	harness	yes, WH-EEG-008	contact-light cable, 10-way

Four coordinates in the table above were wrong until 8 September 2026, and design.py governs (ECO-EEG-016 section 1.4). They are corrected under ECO-EEG-033, which needed them right to write the DXF of fixed connector positions and to lock them in the board file, and found them wrong by comparing every row against the source rather than reading it. **J4** was given as 41.0, 36.0 and is **41.0, 50.0** – ECO-EEG-018 moved it to face its protection rows and this table was not followed through. **J9** was given as 66.0, 78.0 and is **114.0, 62.0**. **J22** was given as 30.0, 116.0 at 90 degrees and is **15.7, 73.5 at 270 degrees** – also an ECO-EEG-018 move, recorded in design.py’s own comment and not here. **J26** was given as 128.0, 104.0 and is **117.0, 8.0**. The two JST footprint names were abbreviated and are given in full. Nothing in design.py moved: this table did. RUL-EEG-021 section C carries the ruling.

Thirty connectors, J1 to J30. Seventeen are module connectors and carry the jumper set of section 3. Two (J6, J7) take the DevKit directly. The remaining eleven are harness, panel, cell and debug, and their cables are specified in WH-EEG-008, not here.

J15, J16 and J17 have no confirmed part. design.py names Staubli SLB1,5-F / LB-I1,5, which is a class of touch-proof 1.5 mm socket rather than a qualified PCB part. A touch-proof 1.5 mm socket with a PCB-mount signal pin and two 1.5 mm retention posts must be sourced and first-articled before Phase 2; AVL-EEG-017 carries a 12-week lead-time risk against it. These are Class A patient-contact parts on every unit, so this is an open item, not a detail.

J22 is not fitted to a panel socket in a standard build. The socket exists on the carrier and both spare channels are protected like every other electrode lead, but the panel DIN sockets, their cable and their drawing are a Phase 2 option and carry no part number (PARTS-EEG-019).

1.1 Pin assignments, module connectors

Every row below is taken from a `conn()` call in `tools/design.py`. Direction is stated from the carrier’s point of view. `NC_*` names are deliberate no-connects: the way exists so that a module which needs the pin can be jumpered without a board change, and it is a single-pad net on the carrier by design.

J1 – ADS1299 module #1, digital (12)

Way	Net	Dir	Domain	Module pin
1	DVDD3V3	out	3.3 V	DVDD
2	DGND	–	0 V	DGND
3	SCLK	out	3.3 V CMOS	SCLK
4	MOSI	out	3.3 V CMOS	DIN
5	MISO	in	3.3 V CMOS	DOUT
6	DRDY	in	3.3 V CMOS	DRDY
7	CS	out	3.3 V CMOS	CS
8	START	out	3.3 V CMOS	START
9	RESET	out	3.3 V CMOS	RESET

Way	Net	Dir	Domain	Module pin
10	CLK_ADS	in	3.3 V CMOS	CLK (module #1 sources the 2.048 MHz clock)
11	V5V	out	5.0 V	module supply input
12	DGND	–	0 V	DGND

J2 – module #1 analogue signals (10) and J23 – module #1 analogue rails (6)

Way	J2 net	Dir	Module pin		Way	J23 net	Dir	Module pin
1	IN1	out	IN1P (Fz)		1	AVDD	in	AVDD
2	IN2	out	IN2P (Cz)		2	AVSS	in	AVSS
3	IN3	out	IN3P (Pz)		3	BIASIN	out	BIASIN
4	IN4	out	IN4P (C3)		4	AGND_REF	–	analogue reference
5	IN5	out	IN5P (C4)		5	AVDD	in	AVDD (second conductor)
6	IN6	out	IN6P (T7)		6	AVSS	in	AVSS (second conductor)
7	IN7	out	IN7P (T8)					
8	IN8	out	IN8P (F7)					
9	SRB1	out	SRB1					
10	BIASOUT	in	BIASOUT					

AVDD and AVSS are **power inputs to the carrier**, produced by module #1's own regulators. The carrier is the load. BIASOUT is likewise an output of the module: it enters the carrier at J2.10, passes through R11 and leaves as BIAS_EL on J14.11 to the Fpz electrode. Note the asymmetry, because it matters to the safety case and not to the interface: R11 is the only one of the sixteen protection resistors that faces outward, and D11 and C11 sit on BIAS_EL, the patient side of it, where the other fifteen have their clamp and filter on the module side. RISK-EEG-011 SF-1a, SF-6a and SR-12 carry the consequence. BIASIN is not driven by anything on the carrier; with the internal channel-average bias loop of E-05 selected it is not required, and the net exists only so a module needing an external feedback return can be strapped at the module end.

J3 – ADS1299 module #2, digital (12)

Way	Net	Dir	Domain	Module pin	Difference from J1
1	DVDD3V3	out	3.3 V	DVDD	–
2	DGND	–	0 V	DGND	–
3	SCLK	out	3.3 V CMOS	SCLK	–
4	MOSI	out	3.3 V CMOS	DIN	–

Way	Net	Dir	Domain	Module pin	Difference from J1
5	DAISY	–	3.3 V CMOS	DOUT	goes to J5.1, not to MISO
6	NC_DRDY2	–	–	DRDY	not used ; TP17 only
7	CS	out	3.3 V CMOS	CS	–
8	START	out	3.3 V CMOS	START	–
9	RESET	out	3.3 V CMOS	RESET	–
10	CLK_ADS	out	3.3 V CMOS	CLK	module #2 receives the clock
11	V5V	out	5.0 V	supply	–
12	DGND	–	0 V	DGND	–

J4 – module #2 analogue signals (10) and J29 – module #2 analogue rails (6)

Way	J4 net	Dir	Module pin		Way	J29 net	Dir	Module pin
1	EMG1	out	IN1P (cheek)		1	AVDD2	in	AVDD
2	EMG2	out	IN2P (submental)		2	AVSS2	in	AVSS
3	EMG3	out	IN3P (laryngeal)		3	BIASIN	out	BIASIN
4	ENV_STIM	out	IN4P		4	AGND_REF	–	analogue reference
5	ENV_VOICE	out	IN5P		5	AVDD2	in	AVDD (second)
6	ENV_ROOM	out	IN6P		6	AVSS2	in	AVSS (second)
7	SPARE1	out	IN7P (EOG)					
8	SPARE2	out	IN8P (EOG)					
9	SRB1	out	SRB1					
10	NC_BIASOUT 2	–	BIASOUT					

J5 – daisy stub to module #1 (4): 1 DAISY (module #2 DOUT into module #1 DAISY_IN), 2 CLK_ADS (module #1 CLKOUT), 3 DGND, 4 DGND. J5 exists because generic ADS1299 breakouts expose DAISY_IN and CLKOUT inconsistently. If module #1's main digital header already carries both, JMP-05 is not fitted and the two nets are picked up on J1.10 instead; record which was done on the build record.

J8 – audio codec module (14) and J9 – microphone feeds (4)

Way	J8 net	Dir	Domain	Codec function
1	DVDD3V3	out	3.3 V	digital and analogue supply

Way	J8 net	Dir	Domain	Codec function
2	DGND	–	0 V	ground
3	I2S_MCLK	out	3.3 V CMOS	master clock in
4	I2S_BCLK	out	3.3 V CMOS	bit clock in
5	I2S_LRCK	out	3.3 V CMOS	word clock in
6	I2S_DIN	in	3.3 V CMOS	codec ADC serial data out
7	I2S_DOUT	out	3.3 V CMOS	codec DAC serial data in
8	SDA	bidir	3.3 V open-drain	control I2C, pulled up on the carrier by R94
9	SCL	out	3.3 V open-drain	control I2C, pulled up on the carrier by R95
10	HP_TAP	in	analogue, <= 1.1 V pk	headphone-amplifier tap
11	HP_L	in	analogue	headphone left out
12	HP_R	in	analogue	headphone right out
13	HP_GND	–	analogue	headphone return
14	V5V	out	5.0 V	headphone amplifier supply

All I2S names are from the **ESP32-S3's** point of view. I2S_DIN is data travelling into the controller and therefore out of the codec. Getting this backwards is the single most common integration error on this interface, and it is why the column above names the codec function separately.

J9: 1 VOICE_PRE (in, analogue), 2 ROOM_PRE (in, analogue), 3 DGND, 4 MIC_MUTE (out, 3.3 V CMOS). Way 4 is present so that a codec module with a hardware ADC-mute input can share the line; if the chosen module has no such pin, way 4 is not mated at the module end and the mute of record is J28.4.

Remaining module connectors

Ref	Way-by-way
J10	1 VDD_ISO (out, 3.3 V via L1) - 2 USB_DN (bidir) - 3 USB_DP (bidir) - 4 DGND
J11	1 DVDD3V3 (out) - 2 DGND - 3 SDA (bidir) - 4 SCL (out)

Ref	Way-by-way
J12	1 VBAT (bidir, cell) - 2 DGND - 3 VBUS_CHG (out, 5 V) - 4 CHG_CE (out, 3.3 V CMOS) - 5 SDA (bidir) - 6 SCL (out) - 7 VSYS (in, 3.0-4.4 V) - 8 NC_CHG_STAT
J19	1 DVDD3V3 (out) - 2 DGND - 3 LED_SR_DATA (out) - 4 LED_SR_CLK (out) - 5 LED_SR_LATCH (out) - 6 LED_OE (out, tied low by R87) - 7 LED_MR (out, pulled up by R88, C88 POR) - 8-15 SR_Q0...SR_Q7 (in) - 16 NC_SR_Q7S
J20	1 DVDD3V3 (out) - 2 DGND - 3 SD_CLK (out) - 4 SD_CMD (bidir) - 5 SD_D0 (bidir) - 6, 7, 8 DGND
J21	1 DVDD3V3 (out) - 2 DGND - 3 VOICE_PRE (in, analogue) - 4 VOICE_RAW (out, analogue) - 5 NC_MIC_GAIN - 6 DGND
J25	1 VSYS (out, 3.0-4.4 V) - 2 DGND - 3 V5V (in, 5.00 V) - 4 DGND - 5 BOOST_EN (out, pulled to VSYS by R86 100 kOhm) - 6 NC_BOOST_PG
J28	1 DVDD3V3 (out) - 2 DGND - 3 ROOM_PRE (in, analogue) - 4 MIC_MUTE (out, 3.3 V CMOS)

1.2 Pin assignments, harness, panel, cell and debug connectors

Ref	Way-by-way
J13	1 VBAT - 2 DGND (JST PH 2.00 mm, keyed)
J14	1 E_Fz - 2 E_Cz - 3 E_Pz - 4 E_C3 - 5 E_C4 - 6 E_T7 - 7 E_T8 - 8 E_F7 - 9 REF_L - 10 REF_R - 11 BIAS_EL - 12 HARN_SHIELD
J15 / J16 / J17	single pole: EMGIN1 / EMGIN2 / EMGIN3
J18	1 VOICE_RAW - 2, 3, 4 DGND
J22	1 EOGIN1 - 2 AGND_REF - 3 EOGIN2
J24	1 VBUS_IN - 2 DGND (JST PH 2.00 mm, keyed)
J26	1 DVDD3V3 - 2 DGND - 3 UART_TX - 4 UART_RX - 5 RESET_EN - 6 NC_GPIO0
J27	1 HP_L - 2 HP_R - 3 HP_GND - 4 NC_HP_DET
J30	1-8 LED1...LED8 - 9 LED_V - 10 LED_GND

Nothing digital enters J14 (ECO-EEG-014). The contact lights leave on J30, a separate cable.

J26 is console and recovery only, and it cannot enter download mode. RFQ-EEG-001 Rev E E-28 asks for TP1 to TP18 plus this 1x6 UART debug header; the 2x5 1.27 mm JTAG/SWD header of Rev C is withdrawn, because the ESP32-S3 has no SWD and its JTAG pins are otherwise used. Way 6 is **NC_GPIO0**, a spare way and not

a boot-mode line: GPIO0 is committed to LED_SR_LATCH at J7 way 14 (ECO-EEG-009). End-of-line and field flashing go through the DevKitC-1's own UART USB-C port, which carries the auto-reset circuit (DTR and RTS to EN and IO0) on the DevKit itself and is reachable through the MP-01 opening of section 4.

2. Required module interfaces

Each module is specified by the signals it must present, not by a brand. A part that meets the interface may be fitted after the qualification of section 6. Two modules are explicitly not substitutable: the ESP32-S3-DevKitC-1-N16R8 (E-18, the pin map and firmware are written to it) and the ADuM4160 isolator (E-24, S-03, it is the safety barrier). RFQ-EEG-001 section 10 additionally names the ADS1299 modules and the ATECC608B breakout as non-substitutable; for those two the **device** is fixed and the breakout board carrying it may be changed under the qualification of section 6, which is the reconciliation AVL-EEG-017 section 2 records.

2.1 ADS1299 eight-channel front-end module, x2 (PIEEG-8 class)

Must present a digital header carrying DVDD (3.3 V in), DGND, SCLK, DIN, DOUT, DRDY, CS, START, RESET and CLK; a 5 V supply input; and an analogue header carrying IN1P-IN8P, SRB1, BIASOUT, BIASIN, AVDD, AVSS and an analogue reference pin. The module must generate AVDD = +2.50 V and AVSS = -2.50 V on board from the 5 V input and must expose both, because the carrier's protection clamps and envelope detectors all run from those rails.

The carrier assumes:

Assumption	Value	Consequence if untrue
AVDD / AVSS available at the header	+/-2.50 V +/-5 %	D1-D16 clamp to the wrong level; U1-U3 have no supply
Spare regulator capacity beyond the converter	>= 25 mA on each rail	carrier analogue load is about 10 mA per rail (<i>calculated: three OPA4376 at 3.04 mA plus the dividers</i>)
Module #1 CLK is an output (CLKOUT enabled)	2.048 MHz	both converters free-run and lose simultaneity (E-01)
Module #2 CLK is an input	–	two drivers contend on CLK_ADS
DAISY_IN reachable, on the header or on J5	–	sixteen channels cannot be daisy-chained
Digital I/O referenced to 3.3 V	V_IH <= 2.48 V, V_OH >= 2.48 V	SPI marginal at >= 4 MHz (E-19)

Assumption	Value	Consequence if untrue
Inputs tolerate the series protection resistance	68 kOhm fitted since ECO-EEG-024 was applied on 2026-09-02; this row read “47 kOhm fitted, 68 kOhm proposed”	E-03 noise budget invalid

The protection network itself is R1-R16, D1-D16 (**BAV99**, not BAT54S – Schottky leakage across the series resistor is an offset error on a 10 uV input) and C1-C16 (10 nF C0G, GCM1885C1H103JA16D). The noise and flatness arithmetic for that network lives in RISK-EEG-011 section 4 and is not recomputed here, and that section has not been re-issued since ECO-EEG-024 was applied, so it still prints the 47 kOhm figures as live.

Corrected 2026-09-02: R1-R16 are 68 kOhm. ECO-EEG-024 is applied in design.py, which gives $2.5\text{ V} / 68\text{ kOhm} = 36.8\text{ uA}$ against S-02’s 50 uA limit, so the limit is met on the calculation; until that date this paragraph said S-02 was not met at the fitted 47 kOhm, at 53.2 uA, and that a module had to tolerate either value. **68 kOhm is the value a module is qualified against from now on.** E-10’s flatness band goes with the resistor: it is **+/-1.0 dB**, the branch that requirement states for exactly the 68 kOhm case, where the 47 kOhm that is no longer fitted sat on the +/-0.5 dB branch. Nothing here has been measured, and **SR-01 is closed in the design and not signed off.**

Which module is #1. The module whose DOUT feeds the other’s DAISY_IN is device 2 in the firmware’s frame order. Module #1 is the one wired to J1/J2/J23 and is the clock source and the rail source. The two modules are not interchangeable once fitted; label both at incoming inspection and keep the pair together. A swap is not caught until TST-EEG-004 T7, where EEG gain is 24 and EMG gain is 12.

[2.2 ESP32-S3-DevKitC-1-N16R8 – see section 5](#)

[2.3 Audio codec module \(ES8388 class\)](#)

Must present an I2S slave port (MCLK, BCLK, LRCK, ADC data out, DAC data in), an I2C control port, at least one stereo DAC output through a headphone amplifier able to drive the shipped headphones to 85 dB SPL at $\leq 0.1\%$ THD (E-13), at least two single-ended ADC inputs, and a node on the headphone-amplifier output usable as HP_TAP.

RFQ A-04 is restated as **32 to 64 Ohm** headphones and the calibrated output level is measured per model. The shipped part is the ATH-M20x at 47 Ohm, so the bench load for the codec measurements is **47.0 Ohm**.

Maximum acoustic output (E-29). The headphone output must not exceed **100 dB SPL** at any commanded level, measured on an artificial ear, and the firmware must clamp the codec volume register to the value measured at calibration. Calculated full-scale output is about 110 dB SPL, which is why the clamp is a requirement and not a preference; a codec module whose amplifier gain cannot be fixed or whose volume register cannot be clamped does not meet this interface. E-29 is verified at **TST-EEG-004 step T28**, a type test run once per lot on one unit and listed in TST-EEG-004 section 14. There is no per-unit acoustic step and this document does not invent one.

HP_TAP is the only pin on this interface that is not a standard module feature. E-16 requires the tap to be taken **after** the amplifier and **before** any user-adjustable control. It must present between 0.1 and 1.1 V peak at the maximum programmed stimulus level, into a source impedance of $\leq 1\text{ kOhm}$. 1.1 V peak is the ceiling because the envelope

chain's output divider (R27/R28, 22 k / 2 k Ω , x0.0909) turns 1.1 V into the +/-100 mV the ADS1299 input sees at gain 1. If the module offers no tap, build one inside the JMP-08 module-end assembly as a 10 k Ω / 1 k Ω divider from HP_L to HP_GND, tap to way 10, and record it. Do not tap before the amplifier.

The codec supply is split: DVDD3V3 on way 1, V5V on way 14 for the headphone amplifier. A module with a single supply pin leaves way 14 unmated and runs from 3.3 V; the 85 dB SPL figure of E-13 must then be re-measured.

2.4 USB isolator module (ADuM4160, not substitutable)

Must present device-side VDD, D-, D+ and GND on a header, with the **host** receptacle on the module itself, presented to the outside world through a gasketed aperture in POD-P1. The host connection is a socket, not a captive cable: WH-08 and the cable gland are deleted from the Phase 1 build and the captive lead through a gland is a Phase 2 item for the helmet shell. The kit ships the two cables of RFQ A-07 and one of them is the host lead. The module must be strapped for full speed (12 Mbit/s) upstream with the device-side pull-up disabled, because the ESP32-S3's native USB PHY provides its own D+ pull-up.

The host connector is a live non-conformance. E-24 asks for USB-C and the named candidate module presents USB-B. The interim resolution is a short USB-B-to-USB-C panel pigtail (**WH-09**) until an isolator module with a USB-C host connector is qualified. That is an interim answer, not a settled interface.

The carrier feeds the device side from DVDD3V3 through L1 (600 Ω at 100 MHz ferrite) as VDD_ISO, bulk-decoupled by C89 (10 μ F) and C87 (100 nF). Isolation $\geq$ 2.5 kV RMS for one minute and creepage $\geq$ 8 mm are S-03 requirements met **on the module**, not on the carrier; that 2.5 kV RMS figure is a type test evidenced by the supplier's certificate at incoming inspection and is never repeated per unit. The per-unit check is the 500 V DC insulation-resistance measurement of TST-EEG-004 step T20.

The carrier's contribution is the isolation keep-out specified in **DSN-EEG-003 section 3.3**, which on a four-layer board means no copper on any of L1, L2, L3 or L4 in that strip – the inner reference planes are cut back from it exactly as the outer layers are. The Rev B DRC report lists no violation of any kind, isolation included, so the strip is clear on all four layers as drawn; that is a report result and not an assumption, and it is the artwork half of the creepage argument REG-EEG-012 clause 8.5.2 rests on. **No conductor of JMP-10 may terminate on any host-side pin of the module, and JMP-10 must not be routed across the module's barrier.** MP-01 is unbroken above the barrier and is printed in PA12 with no conductive filler and no metal insert, so nothing above the plane of the plate bridges it either; the Rev A "20 x 12 mm isolator keep-out slot" was never in mech_gen.py and is withdrawn.

2.5 Secure element breakout (ATECC608B)

Must present VDD (2.0-5.5 V), GND, SDA and SCL. Way count and order on the module are free – the jumper is the adapter. The referenced Adafruit 4314 presents VIN, 3Vo, GND, SCL, SDA, which is a five-way header in a different order from J11's four ways; JMP-11 resolves it (section 3.3). Do not back-feed the module's regulator output (3Vo). The device arrives with its configuration zone unlocked; it is locked and provisioned at end of line (E-21). Swapping this module after provisioning destroys the device identity the browser is bound to.

2.6 Charger and fuel gauge

The baseline is one combined charger-plus-gauge assembly on a single J12 jumper, and JMP-12 is then a straight eight-way jumper. If the programme supplies two separate breakouts, the gauge mounts on MP-01 beside the charger and its VBAT and I2C taps are made at the MP-01 end of the J12 jumper: that is the **Y jumper drawn in section 3.3**, a specified part and not a hand-built variation.

Charger must present VBUS in (4.35-6.0 V), GND, BAT (to the cell), SYS (power-path output), a charge-enable input, and optionally STAT. RFQ E-23 asks for a charger IC with thermal regulation and no charging above 45 C. The module must have the thermal regulation and a candidate without it does not meet this interface, but thermal regulation folds the charge current back on the charger IC's **own die** temperature, which is not the cell's. **E-23's 45 C inhibit is not met and cannot be tested, and no module choice or substitution can close it**: there is no NTC net in design.py and no thermistor way on J12 or J13, so nothing in this design measures cell temperature. **RFQ S-04's thermistor-monitored charging is not met and stays not met** for the same reason. TST-EEG-004 T4 tests the charge interlock and records that the 45 C inhibit is not tested and cannot be; TST-EEG-004 section 10 carries E-23 as not met in part and staying not met; RFQ-EEG-001 Rev E says the same at E-23 and S-04. It is an open hardware item with no closure proposed, listed in DSN-EEG-003 section 11 and RISK-EEG-011, and a module fitted to J12 cannot close it, because the carrier gives the thermistor nowhere to land.

CHG_CE is driven by GPIO47 and holds the charger disabled for the whole of a session (S-01, second interlock; the first is VBUS_DET at J6.14). **Confirm the polarity of the module's enable pin at qualification and record it**; the interlock is a safety function and a wrong-polarity assumption disables it silently.

Gauge must present a cell-sense input, GND, SDA and SCL at 3.3 V logic. ALRT is not brought to the carrier.

2.7 Buck-boost module (TPS63020 class)

Must present VIN, GND, VOUT, GND and an enable input. Output fixed at 5.00 V +/-2 %, >= 1 A continuous, switching above 200 kHz (E-25). Input range must cover the cell across discharge, 3.0 to 4.4 V.

The enable is the trap. R86 pulls BOOST_EN to V5V through 100 kOhm so that V5V comes up before DVDD3V3 exists (ECO-EEG-002). A module carrying its own 10 kOhm pull-down on EN divides that to $3.7 \times 10/110 = 0.34 \text{ V}$ (*calculated*), below any sensible V_IH, and the rail never starts. Measure BOOST_EN with the module fitted and V5V at 3.7 V: it must read >= 1.2 V. A module with an adjustable output must be set to 5.00 V and the trimmer locked with varnish.

V5V load, calculated, not measured. The DevKit's line below is the whole of the DVDD3V3 load of section 5.3, because the DevKit's on-board LDO is fed from V5V at J6.21 and passes very nearly its own output current: an LDO is not a converter, so the 3.3 V load appears on the 5 V rail almost one for one. Rev A's "DevKit about 120 mA" counted the module alone and left the carrier's 3.3 V loads out of the 5 V budget; it is withdrawn, and this table and the table of section 5.3 are now the same arithmetic seen from the two rails.

Load on V5V	Recording, typical (mA)	Worst case simultaneous (mA)
DevKit, including everything its LDO feeds on DVDD3V3 (section 5.3)	190	288
ADS module #1	45	45
ADS module #2	45	45
codec headphone amplifier	20	40 peak
Total on V5V	about 300	about 420

Both columns are inside the 1 A rail. Output ripple limit 50 mV pk-pk measured at TP16 with a 20 MHz-limited probe. This module is a field-replaceable unit and belongs in the SVC-EEG-013 FRU list.

What that means at the cell, and where it disagrees with TST-EEG-004 T3. The buck-boost draws the V5V load from VSYS, so at 3.900 V and an assumed 88 % efficiency the typical figure is $5.00\text{ V} \times 0.300\text{ A} / (3.900\text{ V} \times 0.88) = \text{about } 440\text{ mA at J13}$ while recording, and the worst case is about 610 mA (*calculated*). **TST-EEG-004 T3 limits the recording current at J13 to 150 mA, and that limit and these tables cannot both be right:** on this arithmetic every unit would fail T3. About 440 mA is also what RFQ E-22 allows, since a 3000 mAh cell giving four hours of recording permits 750 mA average. TST-EEG-004 owns the step and its limit, so this document does not restate T3 and does not number a step of its own; the disagreement is carried as open point 13 of section 8 and the first article settles it with a measurement.

2.8 microSD breakout, one-bit SDMMC

Must present a push-push microSD socket wired **directly** to a 3.3 V header: VDD, GND, CLK, CMD and DAT0, with no level shifter and no on-board regulator between the header and the socket. Most “microSD modules” sold for Arduino are SPI adapters with a 5 V regulator and a unidirectional buffer; they cannot work in SDMMC mode, where CMD and DAT0 are bidirectional. This is a hard reject criterion. Card detect is not brought out and firmware must not rely on it.

The sample stream is **50.7 kB/s of frame payload at 1000 Hz** (1015 bytes every 20 ms). E-20’s approximately 70 kB/s is an allowance that covers STATUS and SIGNATURE frames and filesystem overhead; the requirement is not changed by stating the payload figure. The one-bit interface gives about 2 MB/s (*calculated*), which is why the three data lines four-bit mode would have used are spent on the shift register instead (ECO-EEG-009).

2.9 Boom microphone preamplifier

Must present VDD (3.3 V), GND, a single-ended microphone input and a single-ended amplified output. **Gain must be fixed.** E-16’s reference-tone calibration reports a gain, and a gain that moves is not a gain.

Which part is not settled, and it must be treated as not settled. The MAX9814 named in package v1 is an automatic-gain-control part and RFQ E-14 forbids AGC; disabling it is a module-dependent modification, so AVL-EEG-017 keeps the MAX9814 **not approved** and it is named here only as the package-v1 candidate. The preferred

route is a fixed-gain part of the MAX4466 class. Until a part is bought and measured, this module is specified by the interface above and by nothing else. If a candidate implements AGC, the AGC must be disableable by a strap on the module and the strap position recorded; AGC that cannot be disabled disqualifies the module.

Where the preamplifier sits. J21 carries VOICE_RAW **out** to the module and VOICE_PRE **in** from it, and J18 brings VOICE_RAW in from the boom pigtail. The preamplifier is therefore **on MP-01**, and the boom carries the bare electret capsule and its screen. Package v1 described it as being on the boom and WH-EEG-008 Rev B described it as being on the carrier; `design.py` governs, J21 is a carrier socket serving a plate-mounted module, and both of those descriptions are corrected. The boom detaches at the temple for cleaning (E-14) and a detachable preamplifier would be a second thing to lose.

Way 5, NC_MIC_GAIN, is reserved for the module's gain-select pin and is not connected on the carrier. If a strap is needed, make it inside the module-end connector of JMP-21 to way 1 or way 2 and record which.

2.10 Room microphone module

Must present VDD (3.3 V), GND, a single-ended analogue output and a **hardware** mute input that gates the audio path in the analogue domain (E-15). A register-controlled mute does not meet E-15.

MIC_MUTE is defined in this document as **mute-not**: logic high = room microphone live, logic low or open circuit = muted. The net name is retained from `design.py`. GPIO21 is an input at reset and floats, so the muted state must be the module's default with the line open. The carrier carries no pull-down on this net. Verify at qualification: with JMP-28 way 4 unmated and the module powered, the audio path must be attenuated by ≥ 60 dB. If it is not, fit a 100 kOhm pull-down inside the module-end connector of JMP-28 and record it.

No catalogue module is yet known to meet the hardware-mute requirement. If none does, the fallback is an electret capsule with an analogue switch in the module-end connector of JMP-28, which becomes a programme-designed sub-assembly with its own drawing and AVL line. That is recorded in AVL-EEG-017 and is not closed here.

2.11 Shift-register module (74HC595)

Must present SER, SRCLK, RCLK, OE, MR (SRCLR), QA-QH, VCC and GND, all brought to the header, running at 3.3 V. Q0-Q7 sink and source the eight contact lights through R70-R77 (1 kOhm). If the module ties OE or MR internally, ways 6 and 7 are left unmated, R87, R88 and C88 have no effect, and that is recorded on the build record – dark-at-boot does not depend on them.

Contact-light drive, calculated. Bicolour two-lead LED, $V_f = 2.0$ V, series 1 kOhm: $(3.3 - 2.0) / 1000 = 1.3$ mA per site. Eight sites through the LED_V common: **10.4 mA** sourced or sunk by GPIO48 through R78 (0 Ohm), inside the ESP32-S3's 40 mA per-pin limit and well inside the 74HC595's 35 mA per-output and 70 mA per-package limits. LED_V is GPIO48, an input at reset, so nothing can light at boot whatever the shift register holds. LED_MR's power-on reset is $R88 \times C88 = 10 \text{ kOhm} \times 100 \text{ nF} = 1.0 \text{ ms}$ (*calculated*).

The bicolour phase scheme – phase A green, phase B red, alternating to show amber – is specified here as an interface and, **corrected 2026-09-02, it is now implemented in firmware**: until that date `lights_write()` and `lights_task()` were on and off only

and this paragraph said T11 could not pass until they were written. `lights_task()` drives both phases from the converter's positive-side lead-off comparator, read at **two thresholds**. One interface number moves with it: `LIGHT_PHASE_HZ` is still 240, but the half-phase quantises to the FreeRTOS tick, so **the alternation this interface actually carries is about 250 Hz**, which is what E-27's "above 100 Hz" requires and what T11's colorimeter reads. The hardware interface above is unchanged and complete.

Corrected again 2026-09-02 (FW-D17). This paragraph read that the driver took its colours from `LOFF_STATP` and `LOFF_STATN`, and that a build enabling `LOFF_SENSP` only therefore showed amber where it should have shown red. That was true of the code and is no longer, because the premise was wrong: **this interface is single-ended**. J2 carries IN1 to IN8, one shared `SRB1` reference and `BIASOUT`, so there is no per-site negative electrode for `LOFF_STATN` to report on, and enabling `LOFF_SENSN` would have given all eight channels the same bit – the state of the one shared reference. The driver now sweeps `COMP_TH` between a sensitive and an insensitive setting and latches the positive-side result at each: a site that trips neither is **green**, one that trips the sensitive threshold only is **amber**, and one that trips both is **red**. The insensitive set is a subset of the sensitive one by construction, so the three states are exhaustive and cannot overlap. **T11 has not been run**, because no unit exists, and the two `COMP_TH` values are the datasheet's endpoints rather than measured ones – T11 is where they get their real values.

3. The ribbon-jumper schedule

3.1 The jumper part

Property	Specification
Cable	flat ribbon, 28 AWG (7/0.127) stranded tinned copper, PVC, grey with a red stripe on conductor 1. Reference 3M 3365 series, 1.27 mm pitch, used at 2.54 mm by discarding alternate conductors
Analogue jumpers (JMP-02, -04, -23, -29)	screened ribbon; braid or foil with a drain wire, drained at the carrier end only into the <code>AGND_REF</code> crimp of JMP-23.4 / JMP-29.4; module end floating
Carrier end	2.54 mm single-row male contacts (the carrier sockets are female), entering through the WH-KEY-01 shroud
Module end	2.54 mm single-row female contacts (module headers are male)
Way 1	red stripe conductor, both ends, always
Marking	printed heat-shrink sleeve at each end: jumper ID and the connector at that end, e.g. JMP-10 / J10 and JMP-10 / IS0
Maximum length	60 mm , tolerance +0 / -10 mm, measured connector face to connector face

Property	Specification
Minimum bend radius	10 mm
Workmanship	IPC/WHMA-A-620 class 2

Keying is decided. At the **module end**, a 2.54 mm shrouded polarised IDC header is used wherever the module carries one; where the module has only a plain pin strip, pin 1 is marked on the module and the jumper is labelled. At the **carrier end**, the printed keying shroud **WH-KEY-01** – part of the MP-01 print set – fits over the socket and takes the jumper's male contacts, so the jumper can only enter one way round and only on the socket it was made for. Section 6.1 lists which sockets carry a shroud.

The connector system is Harwin M20, and this sentence is a correction. Rev B as first issued specified “a Molex KK 254 crimp build: female housings 22-01-30nn with 08-50-0114 terminals at the module end, free-hanging male housings with 08-52-0072 terminals at the carrier end, with the exact housing part numbers per way count fixed at kitting and listed in AVL-EEG-017”. Three things were wrong with that. AVL-EEG-017 did not carry the per-way-count part numbers, so the pointer was a dead end rather than a stated gap, and thirty-eight connector bodies and about three hundred contacts per unit could not be put on a purchase order. WH-EEG-008 section 6 had already specified **Harwin M20** for the five crimped harness housings, so the kit would have carried two incompatible crimp systems and two crimp tools. And the choice is not free: `tools/mech_gen.py` cuts the WH-KEY-01 shroud cavity at $\text{KEY_CAV_W} = \text{M20_HSG_W} + 0.30 = 4.50$ mm and asserts both that the housing enters with 0.30 mm of clearance and that a reversed housing is stopped by 0.40 mm of interference, so a housing outside **4.20 +0.10 / -0.20 mm across the flats** either will not enter the shroud or will not be keyed by it. A KK 254 body is not that envelope. Taking the Molex wording would therefore have made every printed shroud in the kit the wrong size, silently, and left the one safety-relevant mis-mate in the kit – a reversed WH-01 – with a dust cover in front of it.

The system is therefore **Harwin M20 crimp**: male contacts and polarised male housings at the carrier end, female contacts and housings at the module end, in the way counts of section 3.3 and WH-EEG-008 section 1. **AVL-EEG-017 section 1.6.1 is the single home for the per-way-count housing and contact table, the quantities and the selection criteria**, and this document does not restate them. No Harwin ordering suffix is confirmed: the family roots the programme has quoted are M20-106 for bodies and M20-118 for the male contact, the female contact family has never been named anywhere, and all of them are carried in that section as OPEN WITH CRITERIA to be resolved at the purchase order rather than guessed here.

The qualification is unchanged: 17 jumpers built, every way rung out against the tables of section 1.1, 20 insertion cycles with no contact resistance above 50 mOhm.

3.2 Why 60 mm

The ten-way analogue jumpers carry eight electrode nets, SRB1 and BIASOUT side by side with no interleaved grounds. J2 and J4 have ten ways and every one is a signal; there is no way left to ground. The screen keeps external fields out of the bundle but does nothing between conductors inside it, so length is the only control the design has.

Calculated, with 3M 3365's typical mutual capacitance of 43 pF/m (13 pF/ft):

Step	Value
Coupling capacitance between adjacent conductors at 60 mm	43 pF/m x 0.060 m = 2.6 pF
Victim node shunt (C1-C16 to AGND_REF)	10 nF
Attenuation aggressor to victim	$2.6 \text{ pF} / (2.6 \text{ pF} + 10 \text{ nF}) = 2.6 \times 10^{-4}$ = -71.7 dB
Largest realistic neighbouring artefact (electrode pop, blink)	1 mV pk
Coupled into the neighbour	0.26 uV pk
Front-end noise floor (RISK-EEG-011 section 4, calculated)	0.27 uV RMS

60 mm is the length at which the coupled voltage from the largest artefact a neighbouring electrode is likely to produce falls to the front end's own noise floor. Coupling scales linearly with length: a 120 mm jumper doubles it and puts the artefact 6 dB above the floor. The limit is applied to every jumper, not only the analogue ones, so that the assembler has one number to remember.

E-04 is restated at -80 dB. The Rev C -100 dB crosstalk limit is not achievable through a 60 mm un-interleaved ribbon and, at 40 dB below this instrument's noise floor, is not measurable on this hardware either. E-04 now reads **-80 dB at 50 Hz, measured on the carrier**, with the ribbon's own contribution characterised once on the first prototype (ECO-EEG-026). The -71.7 dB above is the ribbon's calculated contribution and is the honest number for the jumper; it is not the carrier figure E-04 is graded against.

On the digital jumpers 60 mm is not a signal-integrity constraint. Propagation over 60 mm is about 0.3 ns one way (*calculated at 5 ns/m*); a round trip of 0.6 ns is shorter than half the 2 ns edge of a 3.3 V CMOS driver, so no termination is required at the ≥ 4 MHz SPI of E-19 or the 2.048 MHz shared clock.

3.3 Schedule

Jumper	Carrier	Ways	Length (mm)	Module	Screened	Notes
JMP-01	J1	12	55	ADS #1 digital header	no	
JMP-02	J2	10	45	ADS #1 analogue header	yes	drain to JMP-23.4
JMP-23	J23	6	45	ADS #1 rail header	yes	carries the drain of JMP-02
JMP-03	J3	12	55	ADS #2 digital header	no	way 5 is DOUT, way 6 unmated
JMP-04	J4	10	45	ADS #2 analogue header	yes	drain to JMP-29.4
JMP-29	J29	6	45	ADS #2 rail header	yes	see section 7.1 for ways 1, 2, 5, 6
JMP-05	J5	4	40	ADS #1 DAISY_IN / CLKOUT	no	fitted only if needed

Jumper	Carrier	Ways	Length (mm)	Module	Screened	Notes
JMP-08	J8	14	55	codec module	no	ways 10-13 are analogue
JMP-09	J9	4	50	codec ADC inputs	no	way 4 optional
JMP-10	J10	4	40	isolator, device side only	no	must not cross the barrier
JMP-11	J11	4 -> 5	40	secure element	no	adapter jumper, see below
JMP-12	J12	8, or 8 -> 6 + 4	55, branches 25	charger and gauge	no	straight for a combined board, Y for two boards, see below
JMP-19	J19	16	50	74HC595 module	no	
JMP-20	J20	8	45	microSD breakout	no	ways 6-8 all DGND
JMP-21	J21	6	45	boom preamplifier	no	ways 3, 4 analogue
JMP-25	J25	6	45	buck-boost module	no	ways 1, 3 carry rail current
JMP-28	J28	4	40	room microphone	no	

Seventeen jumpers, 134 carrier ways.

JMP-28 and WH-05 are the same connector, and only one of them is built. J28 appears in the schedule above as a 60 mm jumper to the room-microphone module, and in WH-EEG-008 section 3.5 as the carrier end of WH-05, a 180 mm cable to the module on WH-ADP-02. The room-microphone module is **not** on MP-01: WH-EEG-008 section 3.9 bonds WH-ADP-02 to the inside of the POD-P1 wall behind the acoustic port, 180 mm away, so WH-05 is what is actually built and there is no separate 60 mm jumper. AVL-EEG-017 section 1.6.1 counts the 4-way male housing on J28 **once**, and a buyer must not buy two. The row is left in the schedule, and the seventeen-jumper and 134-way figures with it, because ASM-EEG-007 section 8, AVL-EEG-017 section 1.6 and DSN-EEG-003 section 2.1 all quote them into labour estimates and a correction that is not an ECO should not move three documents' arithmetic underneath them. The real count is **sixteen jumpers and 130 carrier ways**, and it is corrected in this schedule at the next revision of this document, under an ECO – open point 15.

ASM-EEG-007 section 8 owns the labour estimate and it is not twenty minutes. Building the set costs 60 minutes per unit in Phase 1, and 25 minutes in Phases 2 and 3 where the set is bought pre-made; fitting it and running the keying check costs a further 20 minutes in every phase (ASM steps 2.4 and 2.6). The “twenty minutes” carried in earlier issues of this section is the size of ASM's fitting step alone and leaves the build out, so it is withdrawn here. **DSN-EEG-003 Rev D section 2.1 still says “roughly twenty minutes of assembly labour” and is wrong by the same margin**; it is corrected at the next revision of that document. Either way the point it was making stands – a hand-built jumper set per unit is the accepted cost of the module architecture, and it is the largest avoidable cost in stage 2.

JMP-11, the adapter jumper. This is the worked example of why the jumper exists.

Carrier way	Net	Module pin (Adafruit 4314)
J11.1	DVDD3V3	VIN

Carrier way	Net	Module pin (Adafruit 4314)
J11.2	DGND	GND
J11.3	SDA	SDA
J11.4	SCL	SCL
–	–	3Vo – not mated

A four-way socket serves a five-way module in a different pin order, with no board change.

JMP-12, the Y jumper. Built only when the charger and the gauge are two separate boards. With the combined assembly of section 2.6 fitted, JMP-12 is a straight eight-way jumper to one module header and the branch columns below are read as one connector.

Trunk way	Net	Charger branch	Gauge branch
1	VBAT	BAT	cell sense
2	DGND	GND	GND
3	VBUS_CHG	VBUS	–
4	CHG_CE	CE	–
5	SDA	–	SDA
6	SCL	–	SCL
7	VSYS	SYS	–
8	NC_CHG_STAT	STAT (optional)	–

Charger branch: six ways (1, 2, 3, 4, 7, 8). Gauge branch: four ways (1, 2, 5, 6). Ways 1 and 2 are common and are made as a single crimp with two conductors, or as two crimps in adjacent positions bridged inside the trunk connector.

3.4 Male headers on the modules

The carrier presents female sockets; the modules must present male pins for the jumper's female end. Several breakouts ship with their header strips loose in the bag.

Module	Header status	Strip needed
ESP32-S3-DevKitC-1-N16R8	fitted by Espressif – confirm the variant ordered	none
PIEEG-8 / ADS1299 breakout x2	confirm part by part at incoming inspection	1x12, 1x10, 1x6 each
codec module	usually loose	1x14 (or the module's own count)
isolator module	usually fitted	1x4

Module	Header status	Strip needed
ATECC608B breakout	loose	1x5
charger module	loose	1x6, or 1x8 for a combined charger-plus-gauge board
fuel gauge module	loose	1x4, none if combined with the charger
microSD breakout	loose	1x8
boom preamplifier	loose	1x6
74HC595 module	loose	1x16
room microphone module	confirm	1x4

Stock: 2.54 mm break-away male header, gold flash over nickel, 1x40 strip – six strips per unit with attrition. Fitted from the component side, soldered on the underside, body flush, perpendicular within 2 degrees. Use the carrier's own socket as the alignment fixture. These lines belong in the kit BOM under “Module interface hardware”; package v1 had none of them.

4. The module plate MP-01

MP-01 is a flat non-conductive plate that sits above the carrier on four standoffs and carries every purchased module except the DevKit. It exists because module header geometry is not fixed by any public standard: nothing tells the programme where a PIEEG-8, an ES8388 breakout or a bq24074 board puts its pins, or how many ways it has, and a socket placed on the carrier to match a guessed geometry is a board respin when the guess is wrong. Moving the modules to a plate makes the variable a jumper and not a PCB.

Every figure below is generated by `tools/mech_gen.py`, which governs, and RUL-EEG-021 section A transcribes the same set.

Property	Value
Material	MJF PA12, natural, no conductive filler and no metal insert
Size	146.0 x 126.0 x 3.0 mm , 2 mm inside the 150.0 x 130.0 mm carrier outline on every edge, corners filleted R4
Board fixing	four Ø3.4 mm holes on the carrier MH1-MH4 pattern, at (5, 5), (145, 5), (5, 125) and (145, 125)
Standoff, carrier to plate	M3 x 18 mm nylon hex, female-female, four off (Würth 970200321 class), with eight M3 x 6 mm nylon pan screws, four up into the plate and four down into the carrier bosses

Property	Value
Jumper slots	a field of slots 12 mm long by 3 mm wide on a 16 x 7 mm grid , inside an 8 mm solid border , deburred; the jumpers drop through the nearest slot to their socket. <code>mech_gen.py</code> cuts each slot with <code>slot2D(12.0, 3.0, 0)</code> , so 12 x 3 mm is the size of one slot and is not a count of slots ; the plate does not carry twelve of them
Module fixing	Ø2.7 mm M2.5 clearance holes between the slot rows , on the same 16 mm column pitch, offset 8 mm in X and 3.5 mm in Y from the slot centres
Module standoff	M2.5 x 6 mm nylon female-female with M2.5 x 5 mm nylon screws; M3 where the module's holes are 3.2 mm
DevKit opening	one 31 x 61 mm opening over J6/J7, spanning x = 78 to 109 mm and y = 4 to 65 mm in carrier coordinates
Above the isolation barrier	solid plate; the slot field stops at x = 140 mm and no material is removed over the barrier
Retention	two adhesive cable-tie mounts per module group

The slot-and-hole field is the answer to the geometry problem. No module's hole pattern is known in advance, so the plate is not drilled to a pattern; any module whose holes fall on or near the M2.5 grid picks up at least two fixings, and one that does not is held by a tie mount and a strip of foam tape. This is deliberately crude and it is honest: it takes a per-unit fitting decision rather than a per-module drawing revision.

Stack budget, calculated, not measured

Element	mm	Running total
POD-P1 floor	2.5	2.5
Carrier boss on the internal floor	6.0	8.5
Carrier PCB	1.6	10.1
Carrier top to plate underside (M3 x 18 standoff)	18.0	28.1
MP-01	3.0	31.1
Modules above the plate, PCB plus tallest component	<= 18.0	49.1
POD-P1 internal depth	55.5	—
Margin	6.4	—

That margin is real and it comes from the enclosure growing with the board: POD-P1 is now 163.0 x 143.0 x 58.0 mm external and 158.0 x 138.0 x 55.5 mm internal. The Rev A figures (a 130 x 124 mm plate in a 142 x 136 x 44 mm pod, margin 0.4 mm) are withdrawn. The 18.0 mm ceiling on module height is a limit on what may be fitted to MP-01, and it is checked at step 9 of section 6 for every module type.

Under the plate, the DevKit occupies 8.5 mm of socket mating height plus a 1.6 mm PCB plus about 3.5 mm of components – **13.6 mm** (*calculated*), leaving 4.4 mm of the 18 mm for jumper bends. The DevKit stands in the 31 x 61 mm opening, so **its USB-C ports and its boot and reset buttons are reachable with the plate fitted**; that is the flashing and recovery route of section 1.2. Every jumper is nonetheless long enough to allow the plate to be lifted and laid beside the carrier without unplugging anything.

No metal above the analogue zone. MP-01 is non-conductive and carries no earthed plate, no metal standoffs and no metal screws left of **x = 62 mm**, the zone split. A conductive plate 18 mm above sixteen high-impedance electrode nodes is a capacitive coupling path to whatever it is tied to, and this design has no earth to tie it to.

5. The ESP32-S3-DevKitC-1 exception

The DevKit's geometry is public, so it is inserted **directly** into J6 and J7. There is no jumper. Row spacing is **22.86 mm (0.900 in)**, the DevKitC-1 header dimension; Rev A had 22.0 mm and the board would not have accepted the module (ECO-EEG-008).

J6 and J7 carry **header positions, not GPIO numbers**. The table below is the reconciliation between the two. It is the one home of the GPIO map – DESIGN_FACTS and FW-EEG-001 cite this section rather than transcribing it – and it is the source from which `firmware/main/board_pins.h` is generated so that the firmware and the board cannot drift.

J6 – DevKitC-1 row A (left header, pin 1 at the USB end)

Way	DevKit label	GPIO	Carrier net	Dir	Note
1	3V3	–	DVDD3V3	out	LDO output, see budget below
2	3V3	–	DVDD3V3	out	second conductor
3	RST	–	RESET_EN	–	to J26.5 only
4	IO4	4	BTN_A	in	R50 pull-up, C50 debounce
5	IO5	5	BTN_B	in	R51, C51
6	IO6	6	BTN_STOP	in	R52, C52
7	IO7	7	I2S_DIN	in	codec ADC data
8	IO15	15	START	out	both converters
9	IO16	16	RESET	out	both converters

Way	DevKit label	GPIO	Carrier net	Dir	Note
10	IO17	17	I2S_MCLK	out	
11	IO18	18	I2S_BCLK	out	
12	IO8	8	I2S_LRCK	out	
13	IO3	3	ENV_CMP	in	strapping pin (JTAG source select); see 5.2 and ECO-EEG-023
14	IO46	46	VBUS_DET	in	strapping pin ; 3.00 V at VBUS = 5 V with R85 at 150 kOhm
15	IO9	9	I2S_DOUT	out	codec DAC data
16	IO10	10	CS	out	
17	IO11	11	MOSI	out	
18	IO12	12	SCLK	out	
19	IO13	13	MISO	in	
20	IO14	14	DRDY	in	interrupt; the sample counter is incremented here and nowhere else
21	5V	–	V5V	in	DevKit supply from the buck-boost
22	GND	–	DGND	–	

J7 – DevKitC-1 row B (right header)

Way	DevKit label	GPIO	Carrier net	Dir	Note
1	GND	–	DGND	–	
2	TXD0	43	UART_TX	out	console, to J26.3
3	RXD0	44	UART_RX	in	console, to J26.4
4	IO1	1	SDA	bidir	shared I2C, pulled up by R94 (4k7)

Way	DevKit label	GPIO	Carrier net	Dir	Note
5	IO2	2	SCL	out	shared I2C, pulled up by R95 (4k7)
6	IO42	42	LED_SR_CLK	out	
7	IO41	41	LED_SR_DATA	out	
8	IO40	40	SD_D0	bidir	
9	IO39	39	SD_CLK	out	
10	IO38	38	SD_CMD	bidir	
11	IO37	37	NC_GPIO37	–	NOT CONNECTED – octal PSRAM
12	IO36	36	NC_GPIO36	–	NOT CONNECTED – octal PSRAM
13	IO35	35	NC_GPIO35	–	NOT CONNECTED – octal PSRAM
14	IO0	0	LED_SR_LATCH	out	strapping pin , pulled up on the DevKit
15	IO45	45	NC_GPIO45	–	NOT CONNECTED – VDD_SPI strapping pin
16	IO48	48	LED_PWM / LED_V	out	input at reset: nothing lights at boot
17	IO47	47	CHG_CE	out	charge interlock (S-01)
18	IO21	21	MIC_MUTE	out	mute-not, see 2.10
19	IO20	20	USB_DP	bidir	native USB
20	IO19	19	USB_DN	bidir	native USB
21	GND	–	DGND	–	
22	GND	–	DGND	–	

5.1 The GPIO35/36/37/45 prohibition

On the **-N16R8** variant, GPIO35, 36 and 37 carry the **octal PSRAM**. Rev A put LED_SR_DATA, LED_SR_CLK and LED_SR_LATCH on them; the firmware pin map was unbuildable and the shift register would have fought the memory bus. ECO-EEG-009 moved them to GPIO41, 42 and 0 and left J7 ways 11, 12 and 13 open. GPIO45 is the VDD_SPI strapping pin and way 15 is also left open. **These four ways carry no carrier copper and no jumper. Nothing may be connected to them by any substitution, rework or field modification.** The four nets exist in the netlist as named single-pad no-connects so that an ERC run reports them as deliberate.

Because four-bit SDMMC is what those lines would have carried, the microSD interface is **one-bit**. The stream needs 50.7 kB/s of payload at 1000 Hz and about 2 MB/s is available (*calculated*), so the trade costs nothing measurable and buys the contact lights.

5.2 Strapping pins at boot

Pin	Way	Function on the carrier	Boot behaviour
GPIO0	J7.14	shift-register latch (RCLK)	pulled up on the DevKit; the 595's RCLK is a high-impedance input, so the strap is not loaded. Pressing BOOT clocks one latch edge, which is harmless because LED_V floats
GPIO3	J6.13	comparator output in	JTAG source select. Driven through R83 (10 kOhm) and clamped by D23; a static level either way boots normally
GPIO45	J7.15	–	open, strap left at its DevKit default
GPIO46	J6.14	charge-present detect	R84 (100 kOhm) / R85 (150 kOhm) divider gives $5 \times 150/250 = 3.00 \text{ V}$ at VBUS = 5 V, above the 2.48 V V_IH, so the first S-01 interlock asserts (ECO-EEG-022)
GPIO48	J7.16	contact-light common	input at reset, so no light can be lit at boot whatever the shift register contains (E-27)

ENV_CMP now arrives from a 3.3 V comparator (ECO-EEG-023). U7 is powered from DVDD3V3 and DGND rather than AVDD and AVSS, so CMP_RAW swings 0 to 3.3 V into GPIO3 with full margin and the D23 clamp becomes a belt-and-braces part rather than the only thing between +/-2.5 V and a 3.3 V pin. Because AGND_REF sits 2.5 V above DGND, the comparator's inputs are re-referenced to a DVDD3V3/2 divider and the envelope is AC-coupled into it. **This change is recorded in ECO-EEG-016 and is**

not yet in design.py, which at Rev B still shows U7 on AVDD/AVSS with its threshold divider referred to AGND_REF. It changes an analogue-to-digital crossing in the patient-connected part of the design and the safety and layout reviewer must check it before it is cut into the board.

5.3 DVDD3V3 comes from the DevKit

There is no 3.3 V regulator on the carrier. DVDD3V3 is the DevKit's own LDO output, taken back onto the board at J6.1 and J6.2, and it feeds every 3.3 V load in the design.

Load	Recording, typical (mA)	Worst case simultaneous (mA)
ADS module #1 digital	10	10
ADS module #2 digital	10	10
codec module, digital and analogue	25	25
secure element	0, idle between sessions	3 peak
74HC595 plus eight lights at 1.3 mA	12	12
microSD breakout	5 between write bursts	100 peak, during write
boom preamplifier	3	3
room microphone module	5	5
isolator device side (VDD_ISO through L1)	20	20
ESP32-S3 itself, radio off	100	100
Total on DVDD3V3	about 190	about 288

U7's 40 uA moves onto this rail **when** ECO-EEG-023 is cut into design.py, which has not happened: at Rev B U7 is still on AVDD and AVSS (section 5.2), so the 40 uA is not on DVDD3V3 yet. It is inside the rounding of the table either way, so no figure in the table moves when the ECO lands. The worst-case column is the 288 mA figure the rest of the package quotes, including TST-EEG-004 T3's note and RUL-EEG-021 section B. Both columns also appear on V5V, because the LDO passes them: section 2.7 carries the 5 V budget and the current at J13 that follows from it, and the two tables are one calculation.

The carrier assumes the DevKit's 3V3 pin will supply 350 mA continuous, and that assumption has not been verified. At 288 mA the on-board LDO drops $5.0 - 3.3 = 1.7$ V and dissipates about **0.5 W** (*calculated*) in a small package inside a closed pod. **This is not solved.** Phase 1 measures it at TST-EEG-004 T3: run the unit at full load, measure DVDD3V3 at TP12, which must stay at or above 3.20 V, and report the regulator's case temperature. If the case exceeds 85 C, a 3.3 V regulator on the carrier fed from V5V is an ECO against Rev C. It is never a field modification.

6. Qualifying a substitute module

Run this before a module type not already in AVL-EEG-017 is fitted to any unit. Record on form QF-EEG-006-01. Steps 1 to 9 are all performed on the bench, on one sample, before the module touches a carrier.

#	Step	Acceptance
1	Obtain the module's schematic and a dimensioned drawing. Confirm the licence permits publishing its pinout, since this package is CC BY-SA.	both obtained
2	Measure header pitch, way count, pin length, pin 1 position and marking.	2.54 +/- 0.10 mm; pin 1 identifiable with the module fitted
3	With the module unpowered, ring out every header pin to its device pin. Write the result into the section 2 table for that module.	every way accounted for, including no-connects
4	Confirm the module's supply range includes the carrier rail it is fed. Measure quiescent current at that rail.	inside the budget of 2.7 or 5.3
5	Measure any on-module regulator output with the carrier rail applied.	recorded; drives the section 7 decisions
6	For every module output that reaches an ESP32-S3 GPIO, measure V _{OH} and V _{OL} under the real load.	V _{OH} >= 2.48 V, V _{OL} <= 0.83 V (0.75 and 0.25 x 3.3 V)
7	For an I2C module, measure its own pull-up to its own VDD, in parallel with the carrier's R94/R95 pair (4k7 each, 2.35 kOhm combined).	recorded; the parallel total must stay above 1.1 kOhm, so that the bus still pulls to <= 0.4 V within the ESP32-S3's 3 mA sink (<i>calculated</i>). If it does not, remove the module's own pull-up
8	For a module on an electrode net, measure leakage from the input pin to any supply at 2.5 V bias.	<= 10 nA, which develops <= 0.7 uV across the 68 kOhm series resistor fitted since ECO-EEG-024 was applied on 2026-09-02 (it was <= 0.5 uV across the 47 kOhm this step used to name as fitted)
9	Measure height above MP-01 and check the hole pattern picks up at least two M2.5 grid holes.	<= 18.0 mm above the plate, inside the section 4 stack budget

#	Step	Acceptance
10	Build the jumper and its WH-KEY-01 shroud, fit, and run TST-EEG-004 in full on one unit.	all steps pass

Sign-off. Phases 1 and 2: the programme's build engineer signs QF-EEG-006-01 and the entry is added to AVL-EEG-017 with an ECO against the affected jumper. Phase 3: the contract manufacturer's engineer signs and the programme countersigns. **No substitution is permitted for the ESP32-S3-DevKitC-1-N16R8 (E-18) or the ADuM4160 isolator (E-24).** A substitution touching the charger, the cell or the isolator additionally requires the electrical safety review that is still outstanding.

6.1 Keying: the socket families and the WH-KEY-01 shroud

The carrier's socket strips repeat: J6 and J7 are both 1x22; J1 and J3 are both 1x12; J2 and J4 are both 1x10; **J21, J23, J25, J26 and J29 are all 1x6; J5, J9, J10, J11, J18, J27 and J28 are all 1x4.** Without a physical key, a 1x4 jumper fits a foreign 1x4 socket and a reversed DevKit across J6 and J7 puts DVDD3V3 onto DGND.

The key is the printed shroud. **WH-KEY-01 is fitted over every socket that takes a ribbon jumper**, which is the seventeen module connectors:

Family	Sockets carrying a WH-KEY-01 shroud
1x16	J19
1x14	J8
1x12	J1, J3
1x10	J2, J4
1x8	J12, J20
1x6	J21, J23, J25, J29
1x4	J5, J9, J10, J11, J28

Each shroud is printed with the socket designator raised on its outer face and is polarised so that a jumper made for a different socket of the same way count will not enter. The three harness sockets **J14, J22 and J30** take the same shroud at the carrier end, specified with the cables in WH-EEG-008 section 6. The SHR-14-A, SHR-30-A and SHR-22-A names of Rev A of JIG-EEG-009 are withdrawn: JIG-EEG-009 Rev C calls all three WH-KEY-01. **J26 (debug), J18 and J27 (panel pigtails), J13 and J24 (JST PH, keyed by the connector itself) and J6/J7 (the DevKit, which is keyed by its own 22-way-by-22.86 mm geometry) carry no shroud** and rely on marking and way count.

Three procedural defences stand behind the shroud and are not withdrawn:

1. **Way 1 is the red stripe** on every jumper, and way 1 of every socket is the square pad with a silkscreen triangle placed outside the socket body so it survives insertion.
2. **Every jumper end carries a printed sleeve** naming both the jumper and the connector.

3. **Where a socket has an NC way** – J3.6, J4.10, J12.8, J19.16, J21.5, J25.6, J26.6, J27.4 and J7.11/12/13/15 – that way is a second key: clip the pin from the jumper's carrier end and plug the socket way with a length of 0.6 mm nylon rod.

The second operator check of ASM-EEG-007 before first power-up remains the last defence.

7. Two build decisions and the measurements that settle them

This section is the only home of the R92/R93 and R89 decisions. Documents that need them cite **ICD-EEG-006 section 7.1** and **section 7.2**. ASM-EEG-007, WH-EEG-008 and ECO-EEG-016 once cited "section 4" and "section 5"; all three now cite 7.1 and 7.2 and that correction is made.

7.1 R92 and R93 – module #2's analogue rails

R92 links AVDD2 to AVDD; R93 links AVSS2 to AVSS. Both are 0 Ohm 0603 and both are **fitted by default**, which parallels module #2's analogue rails onto module #1's regulators (ECO-EEG-011). Removing them lets module #2 run on its own regulators.

Measurement. Fit both modules to MP-01. Fit JMP-01, JMP-03, JMP-23 and JMP-25, but **do not fit JMP-29**. Power the carrier from the cell. Measure at module #2's own AVDD and AVSS header pins, referred to AGND_REF at TP13.

Reading at module #2	Decision	Then
AVDD and AVSS both within +/-100 mV of 0 V	module #2 does not regulate	fit R92 and R93 ; mate JMP-29 with all six ways
module #2 produces +/-2.5 V and $ AVDD1 - AVDD2 \leq 25$ mV and $ AVSS1 - AVSS2 \leq 25$ mV	either is acceptable	keep R92 and R93 fitted; one rail pair, one clamp reference
module #2 produces +/-2.5 V and either difference > 25 mV	the two regulators would fight	remove R92 and R93 ; mate JMP-29 ways 3 and 4 only, and leave ways 1, 2, 5 and 6 unmated

The third case carries a consequence that must be written on the build record: D12-D16 clamp the EMG and EOG inputs to **module #1's** AVDD and AVSS. With R92 and R93 removed, the clamp reference and module #2's supply are different nodes and can differ by the sum of two regulator tolerances. That difference must stay below 100 mV, measured at TP10 and TP11 against module #2's rails, or the clamps sit outside module #2's input range under fault. If it exceeds 100 mV, the only acceptable outcome is to disable module #2's regulators and go back to case one. **There is no numbered production step for that measurement in TST-EEG-004 Rev C.** It is taken at first power-up under this section, recorded on the build record and signed, and the absence of a T-number for a check with a safety consequence is carried as an open item in section 8 rather than fixed by inventing a step number here.

Signed off by the build engineer on QF-EEG-006-01. AGND_REF (J23.4 and J29.4) stays mated in every case; it is the reference every electrode is measured against and it is never a decision.

7.2 R89 – electret bias for the boom capsule

R89 is 2 kOhm 1 % between DVDD3V3 and VOICE_RAW and is **DO NOT POPULATE** by default. It is fitted only if the boom preamplifier module does not supply its own microphone bias. C90 (10 uF) decouples the 3.3 V feed at that point and is fitted either way.

Measurement. Fit the preamplifier module to MP-01 and JMP-21 to J21. Leave J18 unmated, so the capsule is disconnected. Power the carrier. Measure the DC voltage at J21.4 (VOICE_RAW) with a meter of ≥ 10 MOhm input impedance, referred to DGND.

Reading at J21.4	Meaning	Decision
≥ 1.0 V DC	the module biases its own microphone input	leave R89 unpopulated
≤ 0.1 V DC	no bias present	fit R89
between 0.1 and 1.0 V	indeterminate	do not guess: obtain the module schematic and settle it there

Fitting R89 when the module already biases puts two sources on one node. The capsule's DC operating point then sits wrong, the FET clips asymmetrically, and the fault shows up as a distorted VOICE envelope at TST-EEG-004 T12 rather than as anything obviously electrical. Recording the reading is as important as taking the decision. Because the preamplifier part itself is not settled (section 2.9), this measurement is repeated for every candidate module.

With R89 fitted, bias current into a standard electret is $(3.3 - 1.5) / 2200 = \mathbf{0.82\text{ mA}}$ (*calculated*), which is inside the 0.5 to 1.0 mA that most 6 mm capsules want.

8. Open points

Six Rev A open points are closed in this revision and are listed first so that nobody carries them forward: **I2C pull-ups** (R94 and R95, 4k7 each, ECO-EEG-021), **VBUS_DET** (R85 to 150 kOhm, 3.00 V, ECO-EEG-022), **keying** (WH-KEY-01 and the module-end shrouded IDC, section 6.1), **E-04** (restated at -80 dB on the carrier, ECO-EEG-026), **E-11** (C20/C40/C60 to 10 uF, 1.6 Hz, requirement restated at ≤ 2 Hz, ECO-EEG-027) and **E-28** (RFQ-EEG-001 Rev E asks for TP1-TP18 plus the 1x6 header at J26; the 2x5 JTAG header is withdrawn and there is no deviation to record).

What is still open:

#	Point	Evidence	What would close it
1	No safety engineer has reviewed this design , and nothing in it has been built or measured	–	the safety review. It blocks use on a person; it does not block fabrication or quoting

#	Point	Evidence	What would close it
2	J15-J17 have no confirmed part. Staubli SLB1,5-F / LB-I1,5 is a class of touch-proof socket, not a qualified PCB part, on a Class A patient-contact position	section 1; AVL-EEG-017	source a touch-proof 1.5 mm socket with a PCB signal pin and two 1.5 mm posts, and first-article it before Phase 2. AVL carries a 12-week lead-time risk
3	The boom preamplifier part is not settled. The MAX9814 is not approved because its AGC cannot be assumed disableable; the module is specified by interface only	section 2.9; AVL-EEG-017	buy and measure a fixed-gain candidate (MAX4466 class), then add it to AVL-EEG-017 through section 6
4	The isolator's host connector is USB-B where E-24 asks for USB-C. Live non-conformance with an interim answer (WH-09 pigtail), not a settled interface	section 2.4	qualify an isolator module with a USB-C host receptacle, or amend E-24
5	Nothing in this design measures cell temperature. There is no NTC net and no thermistor way on J12 or J13, so S-04's thermistor-monitored charging is not met and E-23's 45 C inhibit is not met and cannot be tested. What the module supplies is the charger IC's own thermal regulation, which regulates its die and not the cell	design.py netlist; TST-EEG-004 T4 and section 10; RFQ-EEG-001 Rev E E-23 and S-04	a carrier change at Rev C, or an accepted deviation. No module fitted to J12 can close it

#	Point	Evidence	What would close it
6	Corrected 2026-09-02: S-02 single-fault DC is 36.8 uA against a 50 uA limit, and the limit is met on the calculation. This item read “53.2 uA ... not met at the fitted 47 kOhm, and it is the one open item of the production simulation SIM-EEG-018”. What is still open is the sign-off: SR-01 is closed in the design and has not been reviewed , which is how SIM-EEG-018 Rev A now carries it among its six open items, and nothing has been measured	RISK-EEG-011 section 4, which still prints the 47 kOhm arithmetic as live; SIM-EEG-018 Rev A	ECO-EEG-024 is applied : R1-R16 are 68 kOhm. E-10 is on its +/-1.0 dB branch , which the requirement provides for this case. The safety reviewer’s disposition of SR-01, and the T23 measurement on a unit that does not yet exist, close it
7	ECO-EEG-023 (ENV_CMP re-reference) is not yet in design.py and touches an analogue-to-digital crossing in the patient-connected part	section 5.2	the safety and layout reviewer signs it, then it is cut into design.py at Rev C
8	The DevKit 3V3 rail is assumed, not measured. 288 mA calculated against an assumed 350 mA, about 0.5 W in the LDO inside a closed pod	section 5.3	TST-EEG-004 T3 at full load: TP12 ≥ 3.20 V and the LDO case temperature reported. Above 85 C, a carrier regulator is an ECO against Rev C
9	The R92/R93 case-three consequence has no numbered test step. A build-time decision with a safety consequence, taken under section 7.1	section 7.1	a step in TST-EEG-004, which owns the step numbers. This document does not number one
10	The EOG panel sockets are a Phase 2 option with no part number. J22 and both channels exist and are protected; the panel sockets, their cable and their drawing do not	section 1; PARTS-EEG-019	a part, a cable and a drawing, if the option is taken

#	Point	Evidence	What would close it
11	Corrected 2026-09-02: the contact-light bicolour phase driver is written, and this item read “not implemented in firmware ... lights_write() and lights_task() are on/off only”. The carrier interface was and is complete. Two things are still open: T11 has not been run, because no unit exists, and the impedance boundaries between green, amber and red are not established. <i>Corrected again the same day (FW-D17): this item read that the build enabled LOFF_SENSP only so the red state could not appear, and asked for LOFF_SENSN to be enabled. Red is reachable now, and LOFF_SENSN was never the answer – the montage is single-ended, so the N half has no per-site electrode. The driver sweeps the positive-side comparator threshold instead, and what is unknown is where the two trip points actually fall</i>	section 2.11; FW-EEG-001	a unit to run T11 on, and the two COMP_TH values characterised so the colour boundaries are known
12	Stack margin is 6.4 mm, calculated. Comfortable, but every figure in it is a drawing dimension and not a measurement	section 4	measure the real stack on the first article before any quantity of MP-01 is printed

#	Point	Evidence	What would close it
13	The rail budgets of sections 2.7 and 5.3 and TST-EEG-004 T3's current limit disagree. About 300 mA on V5V typical implies about 440 mA at J13 while recording; T3 limits the recording current at J13 to 150 mA	sections 2.7 and 5.3, both calculated; TST-EEG-004 T3	the first article: measure idle and recording current at J13 and the five rails, then TST-EEG-004, which owns the step and its limit, sets a limit that matches the measurement. This document does not restate T3
15	J28 is counted twice across the package. Section 3.3 lists it as JMP-28 and WH-EEG-008 section 3.5 lists the same socket as WH-05's carrier end. They are one connector on one socket, and only WH-05 is built, because the room-microphone module is bonded to the pod wall and not fitted to MP-01. The schedule above still says seventeen jumpers and 134 carrier ways	section 3.3; WH-EEG-008 section 3.5 and section 3.9; AVL-EEG-017 section 1.6.1, which counts the housing once	an ECO that deletes JMP-28 from this schedule and restates the count as sixteen jumpers and 130 ways, together with the labour figures ASM-EEG-007 section 8 and DSN-EEG-003 section 2.1 derive from it
14	Rev B was read by a layout engineer, who declined the review and advised a redesign. The DRC was clean against the rule set it had – zero violations, all 145 nets connected, both inner planes continuous – and six of his seven findings are rules that rule set did not contain. Rev B is withdrawn from fabrication; Rev C is unrouted and its placement and routing are bought	ECO-EEG-016 section 2, ECO-EEG-030; the rule set is LAY-EEG-034	the placement review under RFQ-EEG-002A as re-scoped, taken before routing is confirmed. Nothing is released for fabrication

9. Verification and sign-off

Interface	Verified by	When	Signed
Every jumper way, carrier to module	continuity to the tables of sections 1 and 3, 100 %	after jumper build, before first power-up	build operator, checked by a second operator
WH-KEY-01 shroud fitted and correct	visual, against the section 6.1 table	at final assembly	build operator
J6/J7 to GPIO map	firmware/main/board_pins.h generated from section 5 and diffed against it at every build	at every firmware build	automatic; failure breaks the build
Module identity and pinout	QF-EEG-006-01, section 6	before a new module type is fitted	build engineer (Phases 1-2), CM engineer countersigned (Phase 3)
R92 / R93	section 7.1 measurement recorded on the build record	at first power-up of each unit	build engineer
R89	section 7.2 measurement recorded on the build record	at first power-up of each unit	build engineer
DVDD3V3 budget	TST-EEG-004 T3: TP12 at full load, ≥ 3.20 V, with the LDO case temperature reported	first article of each phase	build engineer
Isolation keep-out and jumper routing	visual, against section 2.4 and DSN-EEG-003 section 3.3, on all four layers	at every unit's final assembly	build operator; safety review at the outstanding gate

Nothing in this document has been manufactured or measured, and no safety engineer has reviewed this design. Every figure marked *calculated* is exactly that, and the first article is where they stop being calculations.